

SED SECURITY LOG LAB Assignment_5

Srinivas P Bodke_261100690018

Exercises 1–15 | Authentication Log, Web-Server Log, Configuration Editing and Bash

1. Exercise 1 – Printing Authentication Log Entries

Purpose: print selected entries or ranges from auth.log.

First log entry

```
sed -n '1p' auth.log
```

Explanation: Prints only line 1.

Fifth log entry

```
sed -n '5p' auth.log
```

Explanation: Prints only line 5.

Lines 2–6

```
sed -n '2,6p' auth.log
```

Explanation: Prints the inclusive range from line 2 through line 6.

Last entry

```
sed -n '$p' auth.log
```

Explanation: \$ represents the last line.

Every second entry from line 1

```
sed -n '1~2p' auth.log
```

Explanation: Starts at line 1 and prints every second line.

Failed password entries

```
sed -n '/Failed password/p' auth.log
```

Explanation: Prints only lines containing the specified text.

Successful logins

```
sed -n '/Accepted password/p' auth.log
```

Explanation: Prints successful password-login entries.

First failed login through authentication failure

```
sed -n '/Failed password/,/authentication failure/p' auth.log
```

Explanation: Uses a pattern range: start at the first Failed password and stop at the first authentication failure.

Srinivas P Bodke_261100690018

Exercise 1 – Screenshot Evidence

```
(sri@kali)-[~/Downloads]
$ sed -n '1p' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10

(sri@kali)-[~/Downloads]
$ sed -n '5p' auth.log
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8

(sri@kali)-[~/Downloads]
$ sed -n '2,6p' auth.log
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8

(sri@kali)-[~/Downloads]
$ sed -n '$p' auth.log
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25

(sri@kali)-[~/Downloads]
$ sed -n '1~2p' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30

(sri@kali)-[~/Downloads]
$ sed -n '/Failed password/p' auth.log
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30

(sri@kali)-[~/Downloads]
$ sed -n '/Accepted password/p' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20

(sri@kali)-[~/Downloads]
$ sed -n '/Failed password/,/authentication failure/p' auth.log
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25
```

2. Exercise 2 – Filtering Security Events

Failed login attempts

```
sed -n '/Failed password/p' auth.log
```

Explanation: Selects failed password events.

Successful login attempts

```
sed -n '/Accepted password/p' auth.log
```

Explanation: Selects accepted password events.

Root account entries

```
sed -n '/root/p' auth.log
```

Explanation: Selects lines containing root.

Invalid-user entries

```
sed -n '/Invalid user/p' auth.log
```

Explanation: Selects invalid-user events.

Events for 172.16.0.8

```
sed -n '/172\.16\.0\.8/p' auth.log
```

Explanation: The dots are escaped so they are treated as literal IP-address dots.

Everything except Accepted password

```
sed -n '/Accepted password/!p' auth.log
```

Explanation: ! reverses the match.

Everything except sudo events

```
sed -n '/sudo/!p' auth.log
```

Explanation: Prints lines that do not contain sudo.

Failed password OR Invalid user

```
sed -n -e '/Failed password/p' -e '/Invalid user/p' auth.log
```

Explanation: Uses multiple -e expressions to select either pattern.

Exercise 2 – Screenshot Evidence

```
(sri@kali)-[~/Downloads]
$ sed -n '/Failed password/p' auth.log
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30

(sri@kali)-[~/Downloads]
$ sed -n '/Accepted password/p' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20

(sri@kali)-[~/Downloads]
$ sed -n '/root/p' auth.log
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30

(sri@kali)-[~/Downloads]
$ sed -n '/Invalid user/p' auth.log
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8

(sri@kali)-[~/Downloads]
$ sed -n '/172.16.0.8/p' auth.log
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8

(sri@kali)-[~/Downloads]
$ sed -n '/Accepted password/!p' auth.log
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25

(sri@kali)-[~/Downloads]
$ sed -n '/sudo/!p' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25

(sri@kali)-[~/Downloads]
$ sed -n \
-e '/Failed password/p' \
-e '/Invalid user/p' auth.log
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
```

3. Exercise 3 – Removing Irrelevant Entries

These commands display modified output without changing the original auth.log.

Delete successful logins

```
sed '/Accepted password/d' auth.log
```

Explanation: d deletes matching lines from the output.

Delete Connection closed

```
sed '/Connection closed/d' auth.log
```

Explanation: Removes matching connection-close entries.

Delete first three entries

```
sed '1,3d' auth.log
```

Explanation: Deletes the inclusive line range 1–3.

Delete lines 4–7

```
sed '4,7d' auth.log
```

Explanation: Deletes lines 4 through 7.

Delete 192.168.1.15 entries

```
sed '/192\.168\.1\.15/d' auth.log
```

Explanation: Deletes lines containing that IP address.

Delete every second entry from line 1

```
sed '1~2d' auth.log
```

Explanation: Deletes lines 1, 3, 5, 7 and 9.

Keep only Failed password entries

```
sed '/Failed password/!d' auth.log
```

Explanation: Deletes everything that does not match Failed password.

Exercise 3 – Screenshot Evidence

```
(sri@kali)-[~/Downloads]
$ sed '/Accepted password/d' auth.log
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25

(sri@kali)-[~/Downloads]
$ sed '/Connection closed/d' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25

(sri@kali)-[~/Downloads]
$ sed '1,3d' auth.log
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25

(sri@kali)-[~/Downloads]
$ sed '4,7d' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25

(sri@kali)-[~/Downloads]
$ sed '/192\.168\.1\.15/d' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25

(sri@kali)-[~/Downloads]
$ sed '1~2d' auth.log
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25

(sri@kali)-[~/Downloads]
$ sed '/Failed password/d' auth.log
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
```

4. Exercise 4 – Classifying Security Events

Accepted password

```
sed 's/Accepted password/[SUCCESSFUL LOGIN]/' auth.log
```

Explanation: Replaces the first matching phrase on each line.

Failed password

```
sed 's/Failed password/[FAILED LOGIN]/' auth.log
```

Explanation: Labels failed logins.

Invalid user

```
sed 's/Invalid user/[INVALID USER]/' auth.log
```

Explanation: Labels invalid-user events.

Authentication failure

```
sed 's/authentication failure/[AUTHENTICATION FAILURE]/' auth.log
```

Explanation: Labels authentication failures.

root

```
sed 's/root/PRIVILEGED_USER/' auth.log
```

Explanation: Replaces root with the requested label.

Every server occurrence

```
sed 's/server/authentication-server/g' auth.log
```

Explanation: g replaces every occurrence on each line.

Exercise 4 – Screenshot Evidence

```
(sri@kali)-[~/Downloads]
$ sed 's/Accepted password/[SUCCESSFUL LOGIN]/' auth.log
Sep 10 09:10:12 server sshd[1201]: [SUCCESSFUL LOGIN] for admin from 192.168.1.10
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:13:01 server sshd[1210]: [SUCCESSFUL LOGIN] for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: [SUCCESSFUL LOGIN] for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25

(sri@kali)-[~/Downloads]
$ sed 's/Accepted password/[SUCCESSFUL LOGIN]/' auth.log
Sep 10 09:10:12 server sshd[1201]: [SUCCESSFUL LOGIN] for admin from 192.168.1.10
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:13:01 server sshd[1210]: [SUCCESSFUL LOGIN] for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: [SUCCESSFUL LOGIN] for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25

(sri@kali)-[~/Downloads]
$ sed 's/invalid user/[INVALID USER]/' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: [INVALID USER] hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25

(sri@kali)-[~/Downloads]
$ sed 's/authentication failure/[AUTHENTICATION FAILURE]/' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: [AUTHENTICATION FAILURE] for user student
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25

(sri@kali)-[~/Downloads]
$ sed 's/root/PRIVILEGED_USER/' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:12:15 server sshd[1205]: Failed password for PRIVILEGED_USER from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for PRIVILEGED_USER from 10.0.0.25
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: Failed password for PRIVILEGED_USER from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25

(sri@kali)-[~/Downloads]
$ sed 's/server/authentication-server/g' auth.log
Sep 10 09:10:12 authentication-server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:12:15 authentication-server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 authentication-server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:13:01 authentication-server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:14:33 authentication-server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 authentication-server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:16:11 authentication-server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 authentication-server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:18:05 authentication-server sshd[1235]: Failed password for root from 10.0.0.30
Sep 10 09:20:10 authentication-server sshd[1240]: Connection closed by 10.0.0.25
```

5. Exercise 5 – Masking Sensitive Information

Mask 10.0.0.25

```
sed 's/10\.0\.0\.25/X.X.X.X/g' auth.log
```

Explanation: Replaces that IP address with X.X.X.X.

Mask 172.16.0.8

```
sed 's/172\.16\.0\.8/SUSPICIOUS_IP/g' auth.log
```

Explanation: Replaces the specified IP with SUSPICIOUS_IP.

Mask admin

```
sed 's/admin/USER_A/g' auth.log
```

Explanation: Replaces occurrences of admin.

Mask student

```
sed 's/student/USER_B/g' auth.log
```

Explanation: Replaces occurrences of student.

Replace all digits

```
sed 's/[0-9]/X/g' auth.log
```

Explanation: Replaces each digit with X.

Replace periods with hyphens

```
sed 's/\./-/g' auth.log
```

Explanation: Replaces every literal period in the output.

Redact all IPv4 addresses

```
sed -E 's/[0-9]+\.[0-9]+\.[0-9]+\.[0-9]+/[REDACTED_IP]/g' auth.log
```

Explanation: Uses an extended regular expression to match IPv4-style addresses.

Exercise 5 – Screenshot Evidence (1)

```
(sri@kali)-[~/Downloads]
$ sed 's/10.0.0.25/X.X.X/g' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:12:15 server sshd[1205]: Failed password for root from X.X.X.X
Sep 10 09:12:20 server sshd[1206]: Failed password for root from X.X.X.X
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by X.X.X.X

(sri@kali)-[~/Downloads]
$ sed 's/172.16.0.8/SUSPICIOUS_IP/g' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from SUSPICIOUS_IP
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from SUSPICIOUS_IP
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25

(sri@kali)-[~/Downloads]
$ sed 's/admin/USER_A/g' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for USER_A from 192.168.1.10
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for USER_A from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25

(sri@kali)-[~/Downloads]
$ sed 's/student/USER_B/g' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:13:01 server sshd[1210]: Accepted password for USER_B from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user USER_B
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25
```

Exercise 5 – Screenshot Evidence (2)

```

(sri@kali)-[~/Downloads]
$ sed 's/[0-9]/X/g' auth.log
Sep XX XX:XX:XX server sshd[XXXX]: Accepted password for admin from XXX.XXX.X.XX
Sep XX XX:XX:XX server sshd[XXXX]: Failed password for root from XX.X.X.XX
Sep XX XX:XX:XX server sshd[XXXX]: Failed password for root from XX.X.X.XX
Sep XX XX:XX:XX server sshd[XXXX]: Accepted password for student from XXX.XXX.X.XX
Sep XX XX:XX:XX server sshd[XXXX]: Invalid user hacker from XXX.XX.X.X
Sep XX XX:XX:XX server sshd[XXXX]: Failed password for admin from XXX.XX.X.X
Sep XX XX:XX:XX server sudo[XXXX]: authentication failure for user student
Sep XX XX:XX:XX server sshd[XXXX]: Accepted password for analyst from XXX.XXX.X.XX
Sep XX XX:XX:XX server sshd[XXXX]: Failed password for root from XX.X.X.XX
Sep XX XX:XX:XX server sshd[XXXX]: Connection closed by XX.X.X.XX

(sri@kali)-[~/Downloads]
$ sed 's/\./-/g' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192-168-1-10
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10-0-0-25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10-0-0-25
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192-168-1-15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172-16-0-8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172-16-0-8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192-168-1-20
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10-0-0-30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10-0-0-25

(sri@kali)-[~/Downloads]
$ sed -E 's/[0-9]+\.[0-9]+\.[0-9]+/[REDACTED_IP]/g' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from [REDACTED_IP]
Sep 10 09:12:15 server sshd[1205]: Failed password for root from [REDACTED_IP]
Sep 10 09:12:20 server sshd[1206]: Failed password for root from [REDACTED_IP]
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from [REDACTED_IP]
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from [REDACTED_IP]
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from [REDACTED_IP]
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from [REDACTED_IP]
Sep 10 09:18:05 server sshd[1235]: Failed password for root from [REDACTED_IP]
Sep 10 09:20:10 server sshd[1240]: Connection closed by [REDACTED_IP]

(sri@kali)-[~/Downloads]
$

```

6. Exercise 6 – Filtering Web-Server Requests

Status 200

```
sed -n '/200/p' access.log
```

Explanation: Prints requests containing status 200.

Status 401

```
sed -n '/401/p' access.log
```

Explanation: Prints unauthorized requests in the supplied log.

Status 403

```
sed -n '/403/p' access.log
```

Explanation: Prints blocked requests in the supplied log.

/admin/login

```
sed -n '/\/admin\/login/p' access.log
```

Explanation: Prints requests to the administrative login resource.

POST requests

```
sed -n '/POST/p' access.log
```

Explanation: Prints POST requests.

../requests

```
sed -n '/\.\.\//p' access.log
```

Explanation: Prints requests containing directory-traversal text.

<script> requests

```
sed -n '/<script>/p' access.log
```

Explanation: Prints requests containing the script tag.

OR requests

```
sed -n '/OR/p' access.log
```

Explanation: Prints requests containing OR.

Everything except 200

```
sed -n '/200/!p' access.log
```

Explanation: Prints lines that do not contain 200.

Requests from 10.0.0.25

```
sed -n '/10\.0\.0\.25/p' access.log
```

Explanation: Prints requests from the specified source IP.

Exercise 6 – Screenshot Evidence

```
(sri@kali)-[~/Downloads]
$ sed -n '/200/p' access.log
192.168.1.10 - GET /index.html 200
192.168.1.15 - GET /products.html 200
192.168.1.20 - GET /contact.html 200
192.168.1.25 - GET /about.html 200

(sri@kali)-[~/Downloads]
$ sed -n '/401/p' access.log
10.0.0.25 - GET /admin/login 401
172.16.0.9 - POST /login 401
10.0.0.25 - POST /admin/login 401

(sri@kali)-[~/Downloads]
$ sed -n '/403/p' access.log
172.16.0.8 - GET /search?id=1'OR'1'='1 403
10.0.0.30 - GET /../../etc/passwd 403
172.16.0.8 - GET /script?input=<script> 403

(sri@kali)-[~/Downloads]
$ sed -n '/\admin\login/p' access.log
10.0.0.25 - GET /admin/login 401
10.0.0.25 - POST /admin/login 401

(sri@kali)-[~/Downloads]
$ sed -n '/POST/p' access.log
172.16.0.9 - POST /login 401
10.0.0.25 - POST /admin/login 401

(sri@kali)-[~/Downloads]
$ sed -n '/\.\.\.\.\.p' access.log
10.0.0.30 - GET /../../etc/passwd 403

(sri@kali)-[~/Downloads]
$ sed -n '/<script>/p' access.log
172.16.0.8 - GET /script?input=<script> 403

(sri@kali)-[~/Downloads]
$ sed -n '/<script>/p' access.log
172.16.0.8 - GET /script?input=<script> 403

(sri@kali)-[~/Downloads]
$ sed -n '/OR/p' access.log
172.16.0.8 - GET /search?id=1'OR'1'='1 403

(sri@kali)-[~/Downloads]
$ sed -n '/ 200$/!p' access.log
192.168.1.10 - GET /index.html 200
10.0.0.25 - GET /admin/login 401
172.16.0.8 - GET /search?id=1'OR'1'='1 403
192.168.1.15 - GET /products.html 200
10.0.0.30 - GET /../../etc/passwd 403
172.16.0.9 - POST /login 401
192.168.1.20 - GET /contact.html 200
10.0.0.25 - POST /admin/login 401
172.16.0.8 - GET /script?input=<script> 403

(sri@kali)-[~/Downloads]
$ sed -n '/10\.0\.0\.25/p' access.log
10.0.0.25 - GET /admin/login 401
10.0.0.25 - POST /admin/login 401
```

7. Exercise 7 – Labelling Suspicious Web Requests

200 → [ALLOWED]

```
sed 's/200/[ALLOWED]/g' access.log
```

Explanation: Labels status 200.

401 → [UNAUTHORIZED]

```
sed 's/401/[UNAUTHORIZED]/g' access.log
```

Explanation: Labels status 401.

403 → [BLOCKED]

```
sed 's/403/[BLOCKED]/g' access.log
```

Explanation: Labels status 403.

/admin/login

```
sed 's/\/admin\/login\/\//RESTRICTED_RESOURCE/g' access.log
```

Explanation: Replaces the requested resource path.

GET → READ_REQUEST

```
sed 's/GET/READ_REQUEST/g' access.log
```

Explanation: Labels GET requests.

POST → SUBMIT_REQUEST

```
sed 's/POST/SUBMIT_REQUEST/g' access.log
```

Explanation: Labels POST requests.

Exercise 7 – Screenshot Evidence

```
(sri@kali)-[~/Downloads]
$ sed 's/200/[ALLOWED]/g' access.log
192.168.1.10 - GET /index.html [ALLOWED]
10.0.0.25 - GET /admin/login 401
172.16.0.8 - GET /search?id=1'OR'1'='1 403
192.168.1.15 - GET /products.html [ALLOWED]
10.0.0.30 - GET ../../etc/passwd 403
172.16.0.9 - POST /login 401
192.168.1.20 - GET /contact.html [ALLOWED]
10.0.0.25 - POST /admin/login 401
172.16.0.8 - GET /script?input=<script> 403
192.168.1.25 - GET /about.html [ALLOWED]

(sri@kali)-[~/Downloads]
$ sed 's/401/[UNAUTHORIZED]/g' access.log
192.168.1.10 - GET /index.html 200
10.0.0.25 - GET /admin/login [UNAUTHORIZED]
172.16.0.8 - GET /search?id=1'OR'1'='1 403
192.168.1.15 - GET /products.html 200
10.0.0.30 - GET ../../etc/passwd 403
172.16.0.9 - POST /login [UNAUTHORIZED]
192.168.1.20 - GET /contact.html 200
10.0.0.25 - POST /admin/login [UNAUTHORIZED]
172.16.0.8 - GET /script?input=<script> 403
192.168.1.25 - GET /about.html 200

(sri@kali)-[~/Downloads]
$ sed 's/403/[BLOCKED]/g' access.log
192.168.1.10 - GET /index.html 200
10.0.0.25 - GET /admin/login 401
172.16.0.8 - GET /search?id=1'OR'1'='1 [BLOCKED]
192.168.1.15 - GET /products.html 200
10.0.0.30 - GET ../../etc/passwd [BLOCKED]
172.16.0.9 - POST /login 401
192.168.1.20 - GET /contact.html 200
10.0.0.25 - POST /admin/login 401
172.16.0.8 - GET /script?input=<script> [BLOCKED]
192.168.1.25 - GET /about.html 200

(sri@kali)-[~/Downloads]
$ sed 's/\/admin\/login\/RESTRICTED_RESOURCE/g' access.log
192.168.1.10 - GET /index.html 200
10.0.0.25 - GET /RESTRICTED_RESOURCE 401
172.16.0.8 - GET /search?id=1'OR'1'='1 403
192.168.1.15 - GET /products.html 200
10.0.0.30 - GET ../../etc/passwd 403
172.16.0.9 - POST /login 401
192.168.1.20 - GET /contact.html 200
10.0.0.25 - POST /RESTRICTED_RESOURCE 401
172.16.0.8 - GET /script?input=<script> 403
192.168.1.25 - GET /about.html 200

(sri@kali)-[~/Downloads]
$ sed 's/GET/READ_REQUEST/g' access.log
192.168.1.10 - READ_REQUEST /index.html 200
10.0.0.25 - READ_REQUEST /admin/login 401
172.16.0.8 - READ_REQUEST /search?id=1'OR'1'='1 403
192.168.1.15 - READ_REQUEST /products.html 200
10.0.0.30 - READ_REQUEST ../../etc/passwd 403
172.16.0.9 - POST /login 401
192.168.1.20 - READ_REQUEST /contact.html 200
10.0.0.25 - POST /admin/login 401
172.16.0.8 - READ_REQUEST /script?input=<script> 403
192.168.1.25 - READ_REQUEST /about.html 200

(sri@kali)-[~/Downloads]
$ sed 's/POST/SUBMIT_REQUEST/g' access.log
192.168.1.10 - GET /index.html 200
10.0.0.25 - GET /admin/login 401
172.16.0.8 - GET /search?id=1'OR'1'='1 403
192.168.1.15 - GET /products.html 200
10.0.0.30 - GET ../../etc/passwd 403
172.16.0.9 - SUBMIT_REQUEST /login 401
192.168.1.20 - GET /contact.html 200
10.0.0.25 - SUBMIT_REQUEST /admin/login 401
172.16.0.8 - GET /script?input=<script> 403
192.168.1.25 - GET /about.html 200
```

8. Exercise 8 – Grouping Security Operations

401: replace and print

```
sed -n '/401/{s/401/[UNAUTHORIZED]/;p;}' access.log
```

Explanation: The braces group substitution and printing for matching 401 lines.

403: label, redact IP, print

```
sed -n '/403/{s/403/[BLOCKED]/;s/[0-9]\+\.[0-9]\+\.[0-9]\+\.[0-9]\+/[REDACTED_IP]/;p;}' access.log
```

Explanation: For each 403 line, labels the status, masks the source IP, and prints the result.

admin login: replace URL and IP

```
sed -n -E '/\/admin\/login/{s\/\/admin\/login\/\/PROTECTED_PAGE/g;s/[0-9]+\.[0-9]+\.[0-9]+\.[0-9]\+/[REDACTED_IP]/g;p;}' access.log
```

Explanation: For matching admin-login requests, changes the URL, masks the IP, and prints the result.

Exercise 8 – Screenshot Evidence



```
(sri@kali)~/Downloads
$ sed -n '/401/{s/401/[UNAUTHORIZED]/;p;}' access.log
10.0.0.25 - GET /admin/login [UNAUTHORIZED]
172.16.0.9 - POST /login [UNAUTHORIZED]
10.0.0.25 - POST /admin/login [UNAUTHORIZED]

(sri@kali)~/Downloads
$ sed -n '/403/{s/403/[BLOCKED]/;s/[0-9]\+\.[0-9]\+\.[0-9]\+\.[0-9]\+/[REDACTED_IP]/;p;}' access.log
[REDACTED_IP] - GET /search?id=1'OR'i='1 [BLOCKED]
[REDACTED_IP] - GET ../../etc/passwd [BLOCKED]
[REDACTED_IP] - GET /script?input=<script> [BLOCKED]

(sri@kali)~/Downloads
$ sed -n -E '/ 403$/{s/403/[BLOCKED]/;s/[0-9]+\.[0-9]+\.[0-9]+\.[0-9]\+/[REDACTED_IP]/;p;}' access.log

(sri@kali)~/Downloads
$ sed -n -E '/\/admin\/login/{s\/\/admin\/login\/\/PROTECTED_PAGE/g;s/[0-9]+\.[0-9]+\.[0-9]+\.[0-9]\+/[REDACTED_IP]/g;p;}' access.log
[REDACTED_IP] - GET /PROTECTED_PAGE 401
[REDACTED_IP] - POST /PROTECTED_PAGE 401
```

9. Exercise 9 – Adding Incident Labels

Insert heading

```
sed '1i SECURITY AUTHENTICATION LOG' auth.log
```

Explanation: i inserts text before the first line.

Append ending

```
sed '$a END OF SECURITY LOG' auth.log
```

Explanation: a appends text after the last line.

Warning before failed login

```
sed '/Failed password/i [WARNING]' auth.log
```

Explanation: Inserts a warning before every failed-login line.

Review after invalid user

```
sed '/Invalid user/a Review this event' auth.log
```

Explanation: Appends the review message after matching lines.

Warning before 403

```
sed '/403/i [SUSPICIOUS WEB REQUEST]' access.log
```

Explanation: Inserts the requested label before matching web events.

Investigation after 401

```
sed '/401/a [REQUIRES INVESTIGATION]' access.log
```

Explanation: Appends the investigation label after matching requests.

Replace Connection closed

```
sed 's/. *Connection closed.*/[CONNECTION TERMINATED]/' auth.log
```

Explanation: Replaces the whole matching entry.

Replace successful web requests

```
sed 's/. *200$/[NORMAL WEB REQUEST]/' access.log
```

Explanation: Replaces successful request lines with the requested label when 200 is at the line end.

Exercise 9 – Screenshot Evidence (1)

```
(sri@kali)~(~/Downloads)
$ sed '11 SECURITY AUTHENTICATION LOG' auth.log
SECURITY AUTHENTICATION LOG
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25

(sri@kali)~(~/Downloads)
$ sed '$a END OF SECURITY LOG' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25
END OF SECURITY LOG

(sri@kali)~(~/Downloads)
$ sed '/Failed password/1 [WARNING]' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
[WARNING]
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
[WARNING]
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
[WARNING]
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
[WARNING]
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25

(sri@kali)~(~/Downloads)
$ sed '/Invalid user/a Review this event' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Review this event
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25
```

Exercise 9 – Screenshot Evidence (2)

```
(sri@kali)-[~/Downloads]
$ sed '/ 403$/1 [SUSPICIOUS WEB REQUEST]' access.log
192.168.1.10 - GET /index.html 200
10.0.0.25 - GET /admin/login 401
172.16.0.8 - GET /search?id=1'OR'1'='1 403
192.168.1.15 - GET /products.html 200
10.0.0.30 - GET /../../etc/passwd 403
172.16.0.9 - POST /login 401
192.168.1.20 - GET /contact.html 200
10.0.0.25 - POST /admin/login 401
172.16.0.8 - GET /script?input=<script> 403
192.168.1.25 - GET /about.html 200

(sri@kali)-[~/Downloads]
$ sed '/ 401$/a [REQUIRES INVESTIGATION]' access.log
192.168.1.10 - GET /index.html 200
10.0.0.25 - GET /admin/login 401
172.16.0.8 - GET /search?id=1'OR'1'='1 403
192.168.1.15 - GET /products.html 200
10.0.0.30 - GET /../../etc/passwd 403
172.16.0.9 - POST /login 401
192.168.1.20 - GET /contact.html 200
10.0.0.25 - POST /admin/login 401
172.16.0.8 - GET /script?input=<script> 403
192.168.1.25 - GET /about.html 200

(sri@kali)-[~/Downloads]
$ sed 's/.*Connection closed.*/[CONNECTION TERMINATED]/' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
[CONNECTION TERMINATED]

(sri@kali)-[~/Downloads]
$ sed 's/.* 200$/[NORMAL WEB REQUEST]/' access.log
192.168.1.10 - GET /index.html 200
10.0.0.25 - GET /admin/login 401
172.16.0.8 - GET /search?id=1'OR'1'='1 403
192.168.1.15 - GET /products.html 200
10.0.0.30 - GET /../../etc/passwd 403
172.16.0.9 - POST /login 401
192.168.1.20 - GET /contact.html 200
10.0.0.25 - POST /admin/login 401
172.16.0.8 - GET /script?input=<script> 403
[NORMAL WEB REQUEST]
```

10. Exercise 10 – Separating Log Entries with G

Blank line after every event

```
sed 'G' auth.log
```

Explanation: G appends the hold-space contents; with an empty hold space this produces a blank line.

Blank after every failed login

```
sed '/Failed password/G' auth.log
```

Explanation: Applies G only to failed-login lines.

Blank after every blocked web request

```
sed '/403/G' access.log
```

Explanation: Applies G only to 403 entries.

Blank after every second entry from first

```
sed '1~2G' auth.log
```

Explanation: Applies G to lines 1, 3, 5, 7 and 9.

Exercise 10 – Screenshot Evidence

```
(sri@kali)-[~/Downloads]
$ sed 'G' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25

(sri@kali)-[~/Downloads]
$ sed '/Failed password/G' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25

(sri@kali)-[~/Downloads]
$ sed '/ 403$/G' access.log
192.168.1.10 - GET /index.html 200
10.0.0.25 - GET /admin/login 401
172.16.0.8 - GET /search?id=1'OR'1'='1 403
192.168.1.15 - GET /products.html 200
10.0.0.30 - GET /../etc/passwd 403
172.16.0.9 - POST /login 401
192.168.1.20 - GET /contact.html 200
10.0.0.25 - POST /admin/login 401
172.16.0.8 - GET /script?input=<script> 403
192.168.1.25 - GET /about.html 200

(sri@kali)-[~/Downloads]
$ sed 't=20' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25

(sri@kali)-[~/Downloads]
$
```

11. Exercise 11 – Writing Selected Events to Files

Failed logins

```
sed -n '/Failed password/p' auth.log > failed_logins.txt
```

Explanation: Redirects selected lines to failed_logins.txt.

Successful logins

```
sed -n '/Accepted password/p' auth.log > successful_logins.txt
```

Explanation: Creates successful_logins.txt.

Invalid users

```
sed -n '/Invalid user/p' auth.log > invalid_users.txt
```

Explanation: Creates invalid_users.txt.

401 requests

```
sed -n '/401/p' access.log > unauthorized_requests.txt
```

Explanation: Creates unauthorized_requests.txt.

403 requests

```
sed -n '/403/p' access.log > blocked_requests.txt
```

Explanation: Creates blocked_requests.txt.

172.16.0.8 events

```
sed -n '/172\.\16\.\0\.\8/p' auth.log > suspicious_ip.txt
```

Explanation: Creates suspicious_ip.txt.

Exercise 11 – Screenshot Evidence

```
(sri@kali)~/Downloads
$ sed -n '/Failed password/p' auth.log > failed_logins.txt
(sri@kali)~/Downloads
$ cat failed_logins.txt
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30
(sri@kali)~/Downloads
$ sed -n '/Accepted password/p' auth.log > successful_logins.txt
(sri@kali)~/Downloads
$ cat successful_logins.txt
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from 192.168.1.10
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from 192.168.1.15
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from 192.168.1.20
(sri@kali)~/Downloads
$ sed -n '/Invalid user/p' auth.log > invalid_users.txt
(sri@kali)~/Downloads
$ cat invalid_users.txt
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
(sri@kali)~/Downloads
$ sed -n '/ 401$/p' access.log > unauthorized_requests.txt
(sri@kali)~/Downloads
$ cat unauthorized_requests.txt
(sri@kali)~/Downloads
$ sed -n '/ 401$/p' access.log > unauthorized_requests.txt
(sri@kali)~/Downloads
$ cat unauthorized_requests.txt
10.0.0.25 - GET /admin/login 401
172.16.0.9 - POST /login 401
10.0.0.25 - POST /admin/login 401
(sri@kali)~/Downloads
$ sed -n '/ 403$/p' access.log > blocked_requests.txt
(sri@kali)~/Downloads
$ cat blocked_requests.txt
172.16.0.8 - GET /search?id=1'OR'1='1 403
10.0.0.30 - GET /../etc/passwd 403
172.16.0.8 - GET /script?input=<script> 403
(sri@kali)~/Downloads
$ sed -n '/172\.\16\.\0\.\8/p' auth.log > suspicious_ip.txt
(sri@kali)~/Downloads
$ cat suspicious_ip.txt
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
```

12. Exercise 12 – Combined Security Operations

Label successful and failed logins

```
sed -e 's/Accepted password/[SUCCESSFUL LOGIN]/' -e 's/Failed password/[FAILED LOGIN]/'
auth.log
```

Explanation: Uses two substitutions in one sed invocation.

Label 401 and 403

```
sed -e 's/401/[UNAUTHORIZED]/' -e 's/403/[BLOCKED]/' access.log
```

Explanation: Applies two labels to the web log.

Delete successful, print failed

```
sed '/Accepted password/d' auth.log | sed -n '/Failed password/p'
```

Explanation: First removes successful entries, then prints the remaining failed-login entries.

Replace root and mask IPs

```
sed -E -e 's/root/\[PRIVILEGED_USER\]/g' -e 's/[0-9]+\.[0-9]+\.[0-9]+\.[0-9]+/[REDACTED_IP]/g' auth.log
```

Explanation: Combines account labeling and IP redaction.

GET/POST labels

```
sed -e 's/GET/READ_REQUEST/g' -e 's/POST/SUBMIT_REQUEST/g' access.log
```

Explanation: Labels both HTTP methods.

Exercise 12 – Screenshot Evidence

```
(sri@kali)~[/Downloads]
$ sed -e 's/Accepted password/[SUCCESSFUL LOGIN]/' \
-e 's/Failed password/[FAILED LOGIN]/' auth.log
Sep 10 09:10:12 server sshd[1201]: [SUCCESSFUL LOGIN] for admin from 192.168.1.10
Sep 10 09:12:15 server sshd[1205]: [FAILED LOGIN] for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: [FAILED LOGIN] for root from 10.0.0.25
Sep 10 09:13:01 server sshd[1210]: [SUCCESSFUL LOGIN] for student from 192.168.1.15
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from 172.16.0.8
Sep 10 09:15:42 server sshd[1220]: [FAILED LOGIN] for admin from 172.16.0.8
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: [SUCCESSFUL LOGIN] for analyst from 192.168.1.20
Sep 10 09:18:05 server sshd[1235]: [FAILED LOGIN] for root from 10.0.0.30
Sep 10 09:20:10 server sshd[1240]: Connection closed by 10.0.0.25

(sri@kali)~[/Downloads]
$ sed -e 's/401/[UNAUTHORIZED]/' \
-e 's/403/[BLOCKED]/' access.log
192.168.1.10 - GET /index.html 200
10.0.0.25 - GET /admin/login [UNAUTHORIZED]
172.16.0.8 - GET /search?id=1'OR'1'=1 [BLOCKED]
192.168.1.15 - GET /products.html 200
10.0.0.30 - GET ../../etc/passwd [BLOCKED]
172.16.0.9 - POST /login [UNAUTHORIZED]
192.168.1.20 - GET /contact.html 200
10.0.0.25 - POST /admin/login [UNAUTHORIZED]
172.16.0.8 - GET /script?input=<script> [BLOCKED]
192.168.1.25 - GET /about.html 200

(sri@kali)~[/Downloads]
$ sed -n '/Accepted password/!{/Failed password/p}' auth.log
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30

(sri@kali)~[/Downloads]
$ sed '/Accepted password/d' auth.log | sed -n '/Failed password/p'
Sep 10 09:12:15 server sshd[1205]: Failed password for root from 10.0.0.25
Sep 10 09:12:20 server sshd[1206]: Failed password for root from 10.0.0.25
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from 172.16.0.8
Sep 10 09:18:05 server sshd[1235]: Failed password for root from 10.0.0.30

(sri@kali)~[/Downloads]
$ sed -E \
-e 's/root/\[PRIVILEGED_USER\]/g' \
-e 's/[0-9]+\.[0-9]+\.[0-9]+\.[0-9]+/[REDACTED_IP]/g' auth.log
Sep 10 09:10:12 server sshd[1201]: Accepted password for admin from [REDACTED_IP]
Sep 10 09:12:15 server sshd[1205]: Failed password for [PRIVILEGED_USER] from [REDACTED_IP]
Sep 10 09:12:20 server sshd[1206]: Failed password for [PRIVILEGED_USER] from [REDACTED_IP]
Sep 10 09:13:01 server sshd[1210]: Accepted password for student from [REDACTED_IP]
Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from [REDACTED_IP]
Sep 10 09:15:42 server sshd[1220]: Failed password for admin from [REDACTED_IP]
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: Accepted password for analyst from [REDACTED_IP]
Sep 10 09:18:05 server sshd[1235]: Failed password for [PRIVILEGED_USER] from [REDACTED_IP]
Sep 10 09:20:10 server sshd[1240]: Connection closed by [REDACTED_IP]

(sri@kali)~[/Downloads]
$ sed -e 's/GET/READ_REQUEST/g' \
-e 's/POST/SUBMIT_REQUEST/g' access.log
192.168.1.10 - READ_REQUEST /index.html 200
10.0.0.25 - READ_REQUEST /admin/login 401
172.16.0.8 - READ_REQUEST /search?id=1'OR'1'=1 403
192.168.1.15 - READ_REQUEST /products.html 200
10.0.0.30 - READ_REQUEST ../../etc/passwd 403
172.16.0.9 - SUBMIT_REQUEST /login 401
192.168.1.20 - READ_REQUEST /contact.html 200
10.0.0.25 - SUBMIT_REQUEST /admin/login 401
172.16.0.8 - READ_REQUEST /script?input=<script> 403
192.168.1.25 - READ_REQUEST /about.html 200
```

13. Exercise 13 – security_report.sed

Create the sed script with:

```
nano security_report.sed

1i SECURITY INCIDENT REPORT
s/Failed password/[FAILED LOGIN]/g
s/Invalid user/[INVALID USER ATTEMPT]/g
s/Accepted password/[SUCCESSFUL LOGIN]/g
s/[0-9]\+\.[0-9]\+\.[0-9]\+\.[0-9]\+/[REDACTED_IP]/g
$a END OF INCIDENT REPORT
```

Execute

```
sed -f security_report.sed auth.log
```

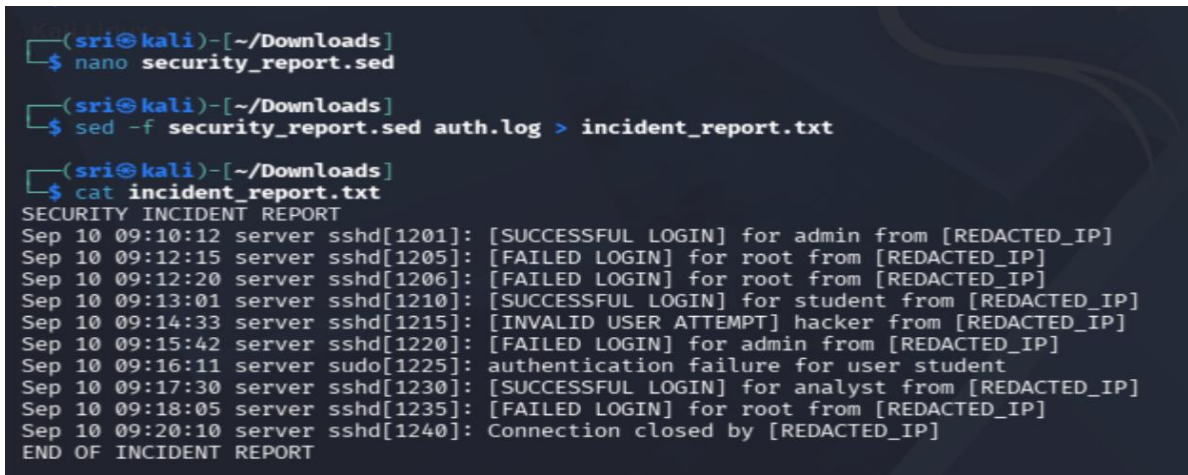
Explanation: -f reads sed commands from the script file.

Save output

```
sed -f security_report.sed auth.log > incident_report.txt
```

Explanation: Redirects the generated incident report to a file.

Exercise 13 – Screenshot Evidence



```
(sri@kali)-[~/Downloads]
$ nano security_report.sed

(sri@kali)-[~/Downloads]
$ sed -f security_report.sed auth.log > incident_report.txt

(sri@kali)-[~/Downloads]
$ cat incident_report.txt
SECURITY INCIDENT REPORT
Sep 10 09:10:12 server sshd[1201]: [SUCCESSFUL LOGIN] for admin from [REDACTED_IP]
Sep 10 09:12:15 server sshd[1205]: [FAILED LOGIN] for root from [REDACTED_IP]
Sep 10 09:12:20 server sshd[1206]: [FAILED LOGIN] for root from [REDACTED_IP]
Sep 10 09:13:01 server sshd[1210]: [SUCCESSFUL LOGIN] for student from [REDACTED_IP]
Sep 10 09:14:33 server sshd[1215]: [INVALID USER ATTEMPT] hacker from [REDACTED_IP]
Sep 10 09:15:42 server sshd[1220]: [FAILED LOGIN] for admin from [REDACTED_IP]
Sep 10 09:16:11 server sudo[1225]: authentication failure for user student
Sep 10 09:17:30 server sshd[1230]: [SUCCESSFUL LOGIN] for analyst from [REDACTED_IP]
Sep 10 09:18:05 server sshd[1235]: [FAILED LOGIN] for root from [REDACTED_IP]
Sep 10 09:20:10 server sshd[1240]: Connection closed by [REDACTED_IP]
END OF INCIDENT REPORT
```

14. Exercise 14 – Modifying security.conf

Initial configuration:

```
PermitRootLogin yes
PasswordAuthentication yes
MaxAuthTries 6
FirewallStatus disabled
LogLevel INFO
GuestLogin enabled
```

Backup

```
cp security.conf security.conf.backup
```

Explanation: Creates a backup before editing.

Apply all changes

```
sed -i \  
-e 's/PermitRootLogin yes/PermitRootLogin no/' \  
-e 's/MaxAuthTries 6/MaxAuthTries 3/' \  
-e 's/FirewallStatus disabled/FirewallStatus enabled/' \  
-e 's/LogLevel INFO/LogLevel VERBOSE/' \  
-e 's/GuestLogin enabled/GuestLogin disabled/' \  
security.conf
```

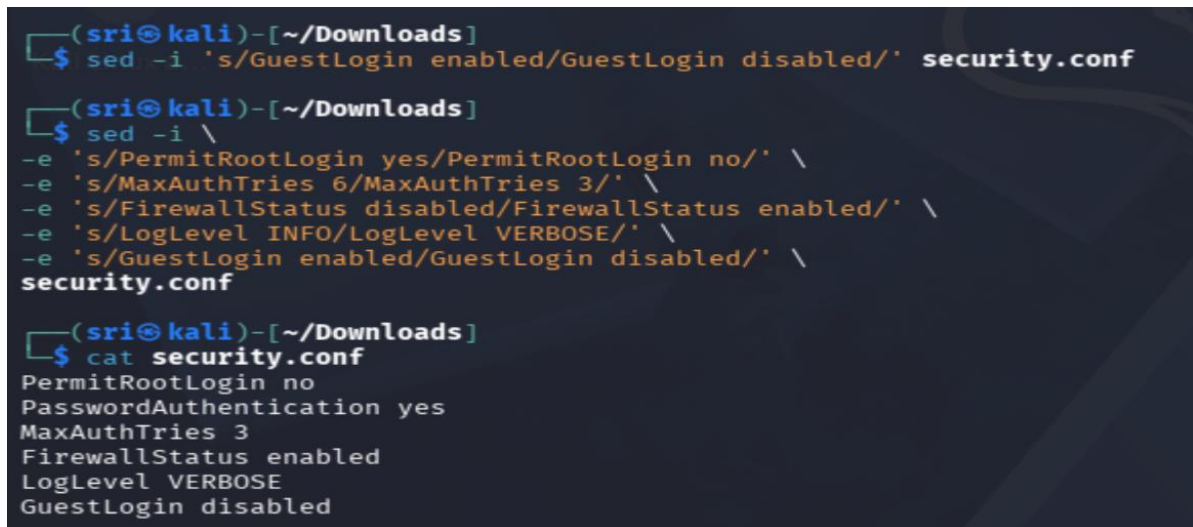
Explanation: Uses -i to edit the file in place and applies five substitutions.

Safer backup form

```
sed -i.bak 's/MaxAuthTries 6/MaxAuthTries 3/' security.conf
```

Explanation: Creates a .bak backup while editing.

Exercise 14 – Screenshot Evidence



```
(sri@kali)-[~/Downloads]  
$ sed -i 's/GuestLogin enabled/GuestLogin disabled/' security.conf  
  
(sri@kali)-[~/Downloads]  
$ sed -i \  
-e 's/PermitRootLogin yes/PermitRootLogin no/' \  
-e 's/MaxAuthTries 6/MaxAuthTries 3/' \  
-e 's/FirewallStatus disabled/FirewallStatus enabled/' \  
-e 's/LogLevel INFO/LogLevel VERBOSE/' \  
-e 's/GuestLogin enabled/GuestLogin disabled/' \  
security.conf  
  
(sri@kali)-[~/Downloads]  
$ cat security.conf  
PermitRootLogin no  
PasswordAuthentication yes  
MaxAuthTries 3  
FirewallStatus enabled  
LogLevel VERBOSE  
GuestLogin disabled
```

15. Exercise 15 – security_log_analyser.sh

Create the Bash script:

```
#!/bin/bash  
  
if [ $# -eq 0 ]; then  
    echo "Usage: $0 <logfile>"  
    exit 1  
fi  
  
if [ ! -f "$1" ]; then  
    echo "Error: File does not exist."  
    exit 1  
fi
```

```

LOGFILE="$1"
OUTPUT="security_alert_report.txt"

echo "SECURITY ALERT REPORT" > "$OUTPUT"
echo "===== " >> "$OUTPUT"

sed -n '/Failed password/p' "$LOGFILE" |
sed -E 's/[0-9]+\.[0-9]+\.[0-9]+\.[0-9]+/[REDACTED_IP]/g' |
sed 's/^/[ALERT] /' >> "$OUTPUT"

sed -n '/Invalid user/p' "$LOGFILE" |
sed -E 's/[0-9]+\.[0-9]+\.[0-9]+\.[0-9]+/[REDACTED_IP]/g' |
sed 's/^/[ALERT] /' >> "$OUTPUT"

echo "END OF REPORT" >> "$OUTPUT"

cat "$OUTPUT"

```

Make executable

```
chmod +x security_log_analyser.sh
```

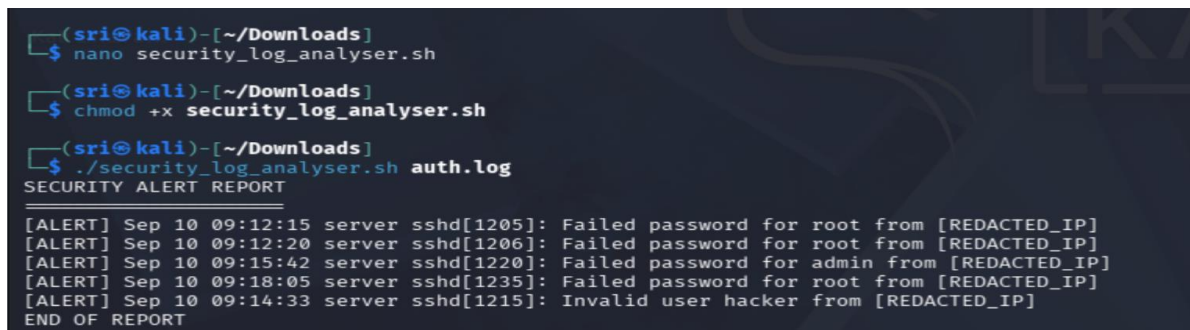
Explanation: Adds execute permission.

Run

```
./security_log_analyser.sh auth.log
```

Explanation: Checks the argument/file, extracts failed and invalid-user events, redacts IPs, labels alerts, saves the report and displays it.

Exercise 15 – Screenshot Evidence



```

(sri@kali)~[~/Downloads]
$ nano security_log_analyser.sh

(sri@kali)~[~/Downloads]
$ chmod +x security_log_analyser.sh

(sri@kali)~[~/Downloads]
$ ./security_log_analyser.sh auth.log
SECURITY ALERT REPORT
[ALERT] Sep 10 09:12:15 server sshd[1205]: Failed password for root from [REDACTED_IP]
[ALERT] Sep 10 09:12:20 server sshd[1206]: Failed password for root from [REDACTED_IP]
[ALERT] Sep 10 09:15:42 server sshd[1220]: Failed password for admin from [REDACTED_IP]
[ALERT] Sep 10 09:18:05 server sshd[1235]: Failed password for root from [REDACTED_IP]
[ALERT] Sep 10 09:14:33 server sshd[1215]: Invalid user hacker from [REDACTED_IP]
END OF REPORT

```